

Freight Rate Prediction

Data review, model comparison, and validation results

1. Data checks

The task is to predict `posted_rate`, the total freight price. The labeled dataset contains 48,000 loads from January through October 2025. The final prediction dataset contains 12,000 loads from November and December.

Issue	Training	Final prediction data
Missing weight	300	165
Weight at or below zero	292	145
Missing market_index	374	249
Duplicate rows	0	0

Zero and negative weights are treated as missing. Missing numerical inputs are filled with training medians, and extra flags mark missing values. Cleaning, scaling, and category encoding are fitted within each training split. `load_id` is excluded from prediction features.

Training prices range from \$57.22 to \$25,533.00. Expensive loads are kept because the supplied files do not establish whether they are errors. Coordinates are used as supplied.

2. Why the split follows the dates

Purpose	Training	Evaluation
Choose models	January-June: 28,806	July-August: 9,671
Final internal test	January-August: 38,477	September-October: 9,523
Final training	January-October: 48,000	Predict 12,000 unlabeled loads

The assessment predicts later months, so the evaluation also uses later months. Model choice is based on July-August MAE. September-October is reserved for reporting final internal performance, without changing the selected models.

3. Models and features

A distance baseline is compared with Ridge regression and histogram gradient boosting. Each model is tested with core features and with market_index and quote_signal added. Models predict dollars per mile, then multiply by distance to obtain the total load price.

Core features include city names, equipment, distance, cleaned weight, four coordinates, weekday, weekend, annual sine and cosine terms, coordinate differences, and log and inverse distance. Full models also use the two supplied signals. Their source is undocumented, so their use assumes they are available when making a prediction.

Ridge uses alpha=50. Boosting uses absolute-error loss, 180 iterations, learning rate 0.07, up to 23 leaves, minimum leaf size 45, L2 regularization 5, and random_state=42. Distance weights make the boosting loss correspond to absolute dollar error. The settings are fixed for this small model comparison.

Candidate	July-August MAE	RMSE
ridge_full	\$148.09	\$625.60
boosting_full	\$148.39	\$630.64
boosting_core	\$152.99	\$626.88
ridge_core	\$188.03	\$634.13
baseline	\$249.01	\$674.72

ridge_full is selected for the main predictions because it has the lowest selection-period MAE. boosting_core is selected for the December chart because that input file does not provide the two extra signals.

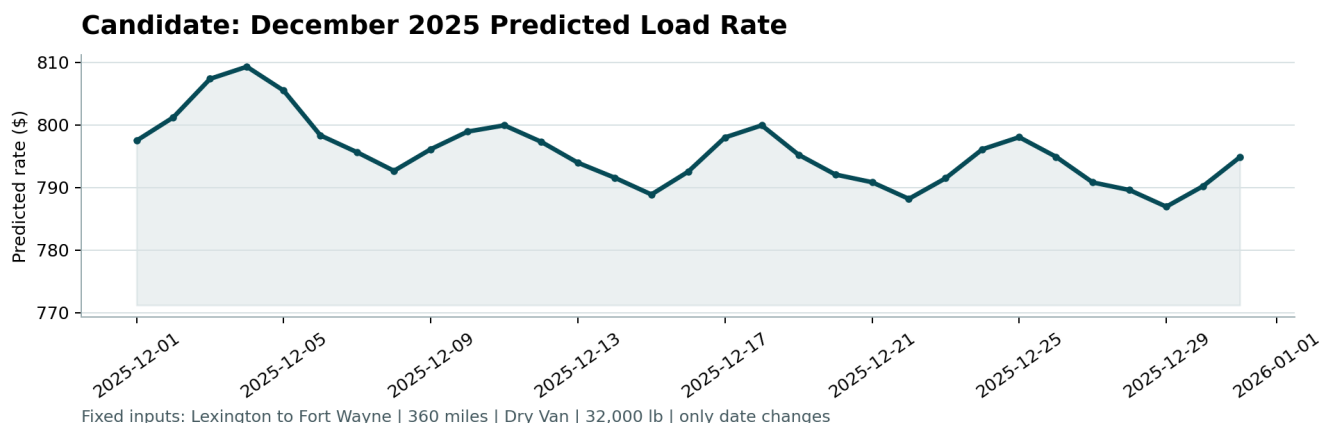
4. Final internal-test results

Model	MAE	RMSE	Median error	MAPE
baseline	\$256.95	\$684.25	\$139.18	11.63%
ridge_full	\$124.28	\$638.13	\$46.59	5.30%
boosting_core	\$119.66	\$637.45	\$43.22	5.16%

The main model has average absolute error of \$124.28, compared with \$256.95 for the baseline. This is a 51.6% improvement. RMSE is \$638.13, which shows that some large errors remain. These are internal-test results; the final November-December accuracy is unknown.

5. December predictions

The December chart keeps Lexington to Fort Wayne, 360 miles, Dry Van, and 32,000 lb fixed. Only the date changes. Coordinates are mapped from the labeled dataset. The core model does not require market_index or quote_signal.



December predictions range from \$786.92 to \$809.29. Changes come from calendar features. Since training contains no November or December examples, the seasonal pattern is uncertain.

6. Output checks and limitations

The original score.py completed successfully. It validates 12,000 load predictions and 31 fixed December predictions, then creates candidate_december.png. This validates the file format, not final prediction accuracy.

Main limitations are expensive outliers, unfamiliar cities, future seasonal behavior, and undocumented signal origins. Further work could test more historical time windows and investigate the largest errors. No unseen-city stress-test result is claimed here because this notebook has not run that experiment.

Source files: train-test.csv, validation.csv, validation-predictions-template.csv, december-chart-inputs.csv, freight-rate-ml-assessment.pdf, and the supplied score.py.